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Problem Set 7
2824 Cryptography homework

Part A: Finding Modular inverses.

1. Find 5 numbers x such that: x * 3 **mod** 10 = 1
   Let x = 7 7 * 3 mod 10 = 21 mode 10 = 1
   Let x = 17 17 * 3 mod 10 = 51 mode 10 = 1
   Let x = 27 27 * 3 mod 10 = 81 mode 10 = 1
   Let x = 37 37 * 3 mod 10 = 111 mode 10 = 1
   Let x = 47 47 * 3 mod 10 = 141 mode 10 = 1

2. Notice that just as we have inverses in multiplication 3 * ⅓ = 1 for example, we can have inverses of 3 **mod** n.

   Are modular inverses unique? Answer why or why not:
   If two integers are relatively prime, the greatest common divisor (gcd) is 1 and an inverse exists. Because the gcd is 1 there is only one solution making the modular inverse unique

   What is the same about each of the x's you found in 1?
   Each of the x's found in part 1 end in the same integer but increment by the value 10 – 7, 17, 27 etc. In this way, each time a value is picked the value * 2 results in a value with the last digit of 1, guaranteeing that when mod 10 is calculated, 1 is the result.

3. See hand written results on following page for problems.
   What is the rule for when Modular inverses exist?
   Modular inverses exist if a and b (a mod b) are relatively prime.

3)

$x \cdot 5 \bmod 10 = 1$

NOT POSSIBLE. 10 is divisible by 5 so mod 10 will always be 0 regardless of x. Also, 5 & 10 are not relatively prime

$x \cdot 2 \bmod 10 = 1$

NOT POSSIBLE. See above.

<u>FIND INVERSES</u>

$x \cdot 7 \bmod 10 = 1$

$x = 3$

$x \cdot 26 \bmod 13 = 1$

NOT POSSIBLE

$x \cdot 7 \bmod 5 = 1$

$x = 3$

$x \cdot 5 \bmod 21 = 1$

$x = 17$

$x \cdot 6 \bmod 35 = 1$

$x = 6$

$x \cdot 7 \bmod 13 = 1$

$x = 2$

Part B:

If an inverse of a mod n exists, what is the gcd(a,n)? Why?

If an inverse exists, the gcd of a and n is 1, therefore the two numbers are relatively prime.

1. through 3. - See handwritten results on the following pages.

4) $\boxed{\text{PART B CONT}}$

$\boxed{\text{Ex 5}}$

a) $a = 4 \quad m = 9$

4 mod 9

$9 = 2 \cdot 4 \boxed{+1} \rightarrow 1 = 9 - 2 \cdot 4$

$4 = 2 \cdot 2$

$2 = 2 \cdot 1$

$\gcd(4, 9) = 1$

$1 = 9 - 2 \cdot 4$

$= \boxed{1} \cdot 9 \boxed{-2} \cdot 4$ $\therefore$ 1 & -2 are Bézout coeffs

$1 \cdot 9 \cancel{\bmod 9} - 2 \cdot 4 \bmod 9 = 1 \bmod 9$

$= -2 \cdot 4 \bmod 9 = 1 \bmod 9$

$\therefore -2$ is inv of 4 mod 9

b) $a = 19 \quad m = 141 \rightarrow$ 19 mod 141

$141 = 7 \cdot 19 + 8$

$19 = 2 \cdot 8 + 3$

$8 = 2 \cdot 3 + 2$

$3 = 1 \cdot 2 \boxed{+1}$

$2 = 2 \cdot 1$

$8 = 141 - 7 \cdot 19$

$3 = 19 - 2 \cdot 8$

$2 = 8 - 2 \cdot 3$

$1 = 3 - 1 \cdot 2$

$\gcd(19, 141) = 1$

$1 = 3 - 1 \cdot 2$

$= 3 - 1 \cdot (8 - 2 \cdot 3) = -1 \cdot 8 + 3 \cdot 3$

$= -1 \cdot 8 + 3 \cdot (19 - 2 \cdot 8) = 3 \cdot 19 - 7 \cdot 8$

$= 3 \cdot 19 - 7 \cdot (141 - 7 \cdot 19) = \boxed{-7} \cdot 141 + \boxed{52} \cdot 19$

-7 & 52 are Bezout coeffs

52 is inv of 19 mod 141

4) PART B CONT

EX 5 CONT

c) $a = 55 \quad m = 89 \quad 55 \bmod 89$

$89 = 1 \cdot 55 + 34 \qquad 34 = 89 - 1 \cdot 55$

$55 = 1 \cdot 34 + 21 \qquad 21 = 55 - 1 \cdot 34$

$34 = 1 \cdot 21 + 13 \qquad 13 = 34 - 1 \cdot 21$

$21 = 1 \cdot 13 + 8 \qquad 8 = 21 - 1 \cdot 13$

$13 = 1 \cdot 8 + 5 \qquad 5 = 13 - 1 \cdot 8$

$8 = 1 \cdot 5 + 3 \qquad 3 = 8 - 1 \cdot 5$

$5 = 1 \cdot 3 + 2 \qquad 2 = 5 - 1 \cdot 3$

$3 = 1 \cdot 2 \boxed{+1} \qquad 1 = 3 - 1 \cdot 2$

$2 \cdot 2 \cdot 1$

$1 = 3 - 1 \cdot 2$

$= 3 - 1 \cdot (5 - 1 \cdot 3) = -1 \cdot 5 + 2 \cdot 3$

$= -1 \cdot 5 + 2(8 - 1 \cdot 5) = 2 \cdot 8 - 3 \cdot 5$

$= 2 \cdot 8 - 3(13 - 1 \cdot 8) = -3 \cdot 13 + 5 \cdot 8$

$= -3 \cdot 13 + 5(21 - 1 \cdot 13) = 5 \cdot 21 - 8 \cdot 13$

$= 5 \cdot 21 - 8 \cdot (34 - 1 \cdot 21) = -8 \cdot 34 + 13 \cdot 21$

$= -8 \cdot 34 + 13(55 - 1 \cdot 34) = 13 \cdot 55 - 21 \cdot 34$

$= 13 \cdot 55 - 21 \cdot (89 - 1 \cdot 55) = \boxed{-21} \cdot 89 + \boxed{34} \cdot 55$

-21 & 34 are Bezout coeffs

34 is inv of 55 mod 89

4) PART B CONT

EX 5 CONT

d) $a = 89$ $m = 232$ 89 mod 232

$232 = 2 \cdot 89 + 54$ $54 = 232 - 2 \cdot 89$

$89 = 1 \cdot 54 + 35$ $35 = 89 - 1 \cdot 54$

$54 = 1 \cdot 35 + 19$ $19 = 54 - 1 \cdot 35$

$35 = 1 \cdot 19 + 16$ $16 = 35 - 1 \cdot 19$

$19 = 1 \cdot 16 + 3$ $3 = 19 - 1 \cdot 16$

$16 = 5 \cdot 3 + \boxed{1}$ $1 = 16 - 5 \cdot 3$

$3 = 3 \cdot 1$

$\gcd(89, 232) = 1$

$1 = 16 - 5 \cdot 3$

$= 16 - 5 \cdot (19 - 1 \cdot 16) = -5 \cdot 19 + 6 \cdot 16$

$= -5 \cdot 19 + 6(35 - 1 \cdot 19) = 6 \cdot 35 - 11 \cdot 19$

$= 6 \cdot 35 - 11(54 - 1 \cdot 35) = -11 \cdot 54 + 17 \cdot 35$

$= -11 \cdot 54 + 17(89 - 1 \cdot 54) = 17 \cdot 89 - 28 \cdot 54$

$= 17 \cdot 89 - 28(232 - 2 \cdot 89)$

$= -28 \cdot 232 + 73 \cdot 89$

-28 & 73 are Bézout coeffs

73 is inv of 89 mod 232

4) $\boxed{\text{PART B CONT}}$

$\boxed{\text{EX 6}}$

a) $a=2 \quad m=17 \qquad 2 \bmod 17$

$17 = 8\cdot 2 + \boxed{1} \qquad 1 = 17 - 8\cdot 2$

$2 = 2\cdot 1$

$1 = 17 - 8\cdot 2$

-8 is inv of $2 \bmod 17$

b) $a=34 \quad m=89 \qquad 34 \bmod 89$

$89 = 2\cdot 34 + 21 \qquad 21 = 89 - 2\cdot 34$

$34 = 1\cdot 21 + 13 \qquad 13 = 34 - 1\cdot 21$

$21 = 1\cdot 13 + 8 \qquad 8 = 21 - 1\cdot 13$

$13 = 1\cdot 8 + 5 \qquad 5 = 13 - 1\cdot 8$

$8 = 1\cdot 5 + 3 \qquad 3 = 8 - 1\cdot 5$

$5 = 1\cdot 3 + 2 \qquad 2 = 5 - 1\cdot 3$

$3 = 1\cdot 2 + 1 \qquad 1 = 3 - 1\cdot 2$

$2 = 2\cdot 1$

$1 = 3 - 1\cdot(5 - 1\cdot 3) = -1\cdot 5 + 2\cdot 3$

$= -1\cdot 5 + 2(8 - 1\cdot 5) = 2\cdot 8 - 3\cdot 5$

$= 2\cdot 8 - 3\cdot(13 - 1\cdot 8) = -3\cdot 13 + 5\cdot 8$

$= -3\cdot 13 + 5(21 - 1\cdot 13) = 5\cdot 21 - 8\cdot 13$

$= 5\cdot 21 - 8(34 - 1\cdot 21) = -8\cdot 34 + 13\cdot 21$

$= -8\cdot 34 + 13(89 - 2\cdot 34) = 13\cdot 89 - 34\cdot 34$

$\boxed{-34}$

4) $\boxed{\text{PART B CONT}}$

$\boxed{\text{EX 6 CONT}}$

c) $a = 144$ $m = 233$ 144 mod 233

$233 = 1 \cdot 144 + 89$ $\quad 89 = 233 - 1 \cdot 144$

$144 = 1 \cdot 89 + 55$ $\quad 55 = 144 - 1 \cdot 89$

$89 = 1 \cdot 55 + 34$ $\quad 34 = 89 - 1 \cdot 55$

$55 = 1 \cdot 34 + 21$ $\quad 21 = 55 - 1 \cdot 34$

$34 = 1 \cdot 21 + 13$ $\quad 13 = 34 - 1 \cdot 21$

$21 = 1 \cdot 13 + 8$ $\quad 8 = 21 - 1 \cdot 13$

$13 = 1 \cdot 8 + 5$ $\quad 5 = 13 - 1 \cdot 8$

$8 = 1 \cdot 5 + 3$ $\quad 3 = 8 - 1 \cdot 5$

$5 = 1 \cdot 3 + 2$ $\quad 2 = 5 - 1 \cdot 3$

$3 = 1 \cdot 2 + 1$ $\quad 1 = 3 - 1 \cdot 2$

$2 = 2 \cdot 1$

$1 = 3 - 1 \cdot 2$

For steps substituting 2 - 13 see ex 6 problem b.

$= -8 \cdot 34 + 13 \cdot 21$

$= -8 \cdot 34 + 13 \cdot (55 - 1 \cdot 34) = 13 \cdot 55 - 21 \cdot 34$

$= 13 \cdot 55 - 21(89 - 1 \cdot 55) = -21 \cdot 89 + 34 \cdot 55$

$= -21 \cdot 89 + 34 \cdot (144 - 1 \cdot 89) = 34 \cdot 144 - 55 \cdot 89$

$= 34 \cdot 144 - 55 \cdot (233 - 1 \cdot 144) = -55 \cdot 233 + 89 \cdot 144$

$\boxed{89}$ is inv of 144 mod 233

PART B CONT

EX 6 CONT

d) $a = 200 \quad m = 1001 \qquad 200 \bmod 1001$

$1001 = 5 \cdot 200 + 1 \qquad 1 = 1001 - 5 \cdot 20$

$200 = 200 \cdot 1$

$\boxed{-5}$ is in $200 \bmod 1001$

EX 33

$7^{121} \bmod 13$

$= 7^{12 \cdot 10 + 1} \bmod 13$

$= (7^{12})^{10} \cdot 7^{1} \bmod 13$

$= (7^{12})^{10} \bmod 13 \cdot 7^{1} \bmod 13$

$= (1)^{10} \bmod 13 \cdot 7 \bmod 13$

$= 1 \cdot 7 \bmod 13$

$= \boxed{7}$

EX 34

$23^{1002} \bmod 41$

$= 23^{40 \cdot 25 + 2} \bmod 41$

$= (23^{40})^{25} \bmod 41 \cdot 23^{2} \bmod 41$

$= (1)^{25} \bmod 41 \cdot 529 \bmod 41$

$= 1 \cdot 37$

$= \boxed{37}$

2. Pseudocode for the Extended Euclidean Algorithm (adapted from Rosen KH, Discrete Mathematics)

```
def Ext_Eucl(a, b):  // input variables are a and b where a and b are integers and in the form
                     // a mod b
    // example: a mod b = 4 mod 9
    // initialization section
    // Euclideans Algorithm for gcd is in the form
    // 9 = 2 * 4 + 1          9 = m, 2 = s, 4 = q , 1 = 4
    // 4 = 2 * 2 + 0
    // Ext. Eucl Alg
    // 1 = 9 – 2*4            1 = m, x = 9, s = 2, q = 4
    x = 0            // set variable to store first Bezout coefficient
    y = 1            // set variable to store second Bezout coefficient
    s = 1            // set variable for multiplier of first Bezout coefficient
    t = 0            // set variable for multiplier of first Bezout coefficient

    // main section
 while a != 0:       // the program will terminate with the divisor is == 0
    q = b // a       // calc the quotient of b divided by a
    r = b % a        // calc the remainder of b divided by a
    m = x – s * q    // calc ext eucl of x
    n = y – t * q    // calc ext eucl of y
    // increment previously defined variables
    b = a            // set new divisor to a
    a = r            // set a to remainder
    x = s            // set bez coeff to multiplier
    y = t            // set bez coeff to multiplier
    s = m            // set multiplier to ext euc of x
    t = n            // set multiplier to ext euc of y

gcd = b              // once while loop is exited, gcd = b
   return gcd, x, y  // return gcd and two Bezout coefficients x and y
```

Part C:

Example 8

Message to encrypt: STOP

Keys: (2537, 13) in the form (n,e) where n = p * q where p = 43 and q = 59 (both of which are primes) and e is a selected prime number that is < n and != p or q.

Test that e and (p-1)(q-1) are relatively prime:

$\gcd(e, (p-1)(q-1)) = \gcd(13, 42 \cdot 58) = 1.$

1) Encrypt characters in STOP to numerical equivalents using two digit representations (the value of STOP then becomes 18191415 where S = 18 and P = 15. The underlying "shift cypher" is each character A through Z is incremented by 9 sets of doubles (A = 00, B = 01, K = 00, L = 11 etc).
2) Divide the message into blocks of 4 integers.
3) Encrypt each block using the form $C = M^{13} \bmod 2537$ where M is the two blocks of message to be encoded using fast modular multiplication.
   $1819^{13} \bmod 2537 = 2081$
   $1415^{13} \bmod 2537 = 2182$
4) The resulting encrypted message is 20812182

Example 9

Message received: 0981 0461

Decrypt using the same values as in example 8 – p = 43, q = 53, e = 13

1) Calcuate the inverse of 13 mod (p-1)(q-1) this is then the value of d
   13 mod 42*58 = 13 mod 2436
   d*13 mod 2436 = 1
   d = 937
2) Calculate the following
   $M = C^{937} \bmod 2436$ where m is the decrypted message and C is the tw values that were received
   $M = 0981^{937} \bmod 2436 = 0704$
   $M = 0461^{937} \bmod 2436 = 1115$
3) Decrpyted message is HELP (07-04-11-15)